

StocksDeepDive — Complete User Manual

stocksdeepdive.com · Version of 13 August 2026 · supersedes "Global Stock Scanner — Calculation Flow"

This manual describes every page, every table, every score and every colour on the site, exactly as deployed. Nothing here is left to interpretation: each formula is written out, each threshold is stated, and each colour is defined. It replaces the earlier "Global Stock Scanner — Calculation Flow" document, which described the desktop application; several features documented there (Strategy Mode selector, Swing Trading mode, Auto-Trading/Alpaca, Mega Cap and Turnaround universes) are **not part of the public website**.

1. Site structure and navigation

The site has eight addressable pages:

URL	Page	Purpose
/	Home	Landing page: search, featured analysis, market strip, feature overview
/deep-dive	Stock Deep Dive	Complete analysis of one ticker
/comparison	Side-by-side Comparison	Two or more tickers on identical criteria
/scanner	Stock Scanner	Rank a whole index (ASX 200, S&P 500, etc.)
/research	Rational Compounder Analysis	The author's hand-built workbook research
/methodology	How the scores work	Plain-English methodology reference
/about	About	Who runs the site and why
/privacy	Privacy policy	What data the site handles

The search box is the primary navigation. It appears at the top of every page. The rule is simple and absolute: type **one** ticker and press Search (or Enter) for a Deep Dive; type **two or more** tickers — separated by spaces, commas or new lines — for a Comparison. ASX tickers use the `.AX` suffix (`CSL.AX`); US tickers use none (`AAPL`); the two can be mixed freely in one comparison. Duplicate tickers are ignored, case does not matter.

Below the search box sit **three buttons**: *Rational Compounder Analysis* (left), *Side-by-side Comparison* (middle) and *Stock Scanner* (right). These open the pages directly — Comparison opens showing its last results if you ran one this session, otherwise an empty state with one-click example chips.

Example chips. On the home page and on Deep Dive/Comparison empty states, clickable chips (`CSL.AX` , `AAPL` , `BHP.AX` , `RMD.AX` , `CSL.AX vs BHP.AX`) run that exact search with one click.

Shareable URLs. Results pages keep the address bar in sync with what is on screen.

`stocksdeepdive.com/deep-dive?ticker=CSL.AX` opens directly into a live CSL analysis;
`stocksdeepdive.com/comparison?tickers=CSL.AX,BHP.AX` runs that comparison directly. Copying the address bar at any time captures the current view — these links can be embedded in blog posts, emails or messages.

Clicking the StocksDeepDive logo returns to the home page from anywhere.

2. The home page

From top to bottom:

Market tape. A scrolling strip of live quotes (ASX 200, S&P 500, NASDAQ, AUD/USD, plus sample stocks) with today's percentage move — green ▲ for up, red ▼ for down. Refreshed every 30 minutes. If quotes cannot be fetched, the tape simply doesn't render.

Account bar. Sign In (Google) when signed out; your name, a feedback button and Sign out when signed in. See §9 for what signing in adds.

Hero and search. The headline, the search box with an Analyze button, the example chips, and the usage rule ("One ticker = full Deep Dive · Two or more = side-by-side Comparison").

Featured analysis card. A live, attention-lite Deep Dive of one stock, rotating daily through a fixed list (CSL.AX, AAPL, BHP.AX, RMD.AX, MSFT, WES.AX, GOOGL) and cached for six hours. It shows the Long Score on a colour-coded gauge with its verdict, intrinsic value, margin of safety, Quality Score, verdict pills (valuation / sentiment / entry), and a 6-month price sparkline against the 50-day average. The button beneath opens the full Deep Dive. If the analysis cannot be fetched, an explainer card renders instead.

Your watchlist (signed-in users with saved stocks only): a chip row of your saved tickers — one click re-runs any of them.

Market mood strip. Up to four tiles: the AU and US market mood (Hopeful / Neutral / Anxious — a rolling 10-day average of GDELT's news-tone reading for that country; informational only, it feeds no score) and index levels with today's move. A tile that can't be fetched (e.g. GDELT rate-limiting) is silently omitted and returns when the source recovers.

Feature cards, How-it-works, coverage cards and the launch banner complete the page. All remote data on the home page is fetched concurrently with a 10-second budget — the page itself always renders immediately; a slow source only delays its own box.

3. The scoring engines (used identically everywhere)

Every tool on the site runs the same engine. All live data comes from Yahoo Finance (prices, volumes, fundamentals, analyst estimates), Google Trends (search interest), NewsAPI + Yahoo news (headlines) and StockTwits (social chatter).

3.1 Quality Score (0–100)

"Is this a good business?" — computed from the company's own fundamentals:

$$\text{Quality} = 50 + \text{ROIC} \times 100 \times 0.25 + \text{ROE} \times 100 \times 0.20 + \text{Profit Margin} \times 100 \times 0.15 + \text{Revenue Growth} \times 100 \times 0.15 + \text{Earnings Growth} \times 100 \times 0.15 + 10 \text{ if free cash flow positive } (-5 \text{ if negative}) - \min(\text{Debt/Equity} \times 0.05, 15) \text{ [debt penalty, capped]}$$

- **ROIC** (return on invested capital) is the lead metric — it measures profit per dollar of *total* capital (equity **and** debt), so a business cannot look high-quality on ROE alone by borrowing heavily. It is derived as: book equity = market cap ÷ price-to-book; ROIC ≈ net income ÷ (book equity + total debt); clamped to ±100%. When the inputs are unavailable the term contributes zero.
- Revenue and earnings growth are **clamped to ±50%** before weighting, so a +300% figure off a tiny base cannot max the score.

- **Profitability gate:** a loss-making or cash-burning business (negative margin, net income or free cash flow) is **capped at 55** regardless of growth.
- The final score is clamped to 0–100. If no fundamental data was available at all, the score defaults to 50 and is rendered **in red** (see the red-flag rule, §3.7).

The Deep Dive's "What's driving Quality" chart shows each of these terms as a bar.

3.2 Intrinsic value and Margin of Safety

"What is it worth?" The valuation ladder, in order:

1. **DCF** — used whenever free cash flow is positive. Growth = analyst consensus (Yahoo 5-year estimate) → else the company's own capex-normalised historical FCF growth → else reported growth → else a flagged default. Discount rate = per-stock **CAPM** (the stock's own beta against its market's risk-free rate and premium). Terminal growth is set by the stock's currency.
2. **P/E blend** fallback when a DCF isn't possible — labelled as such in the "Val Method" fields.
3. **N/A** when neither works (typically financials, or no positive EPS/FCF).

$$\text{Margin of Safety (MOS)} = (\text{Intrinsic Value} - \text{Price}) / \text{Intrinsic Value} \times 100$$

Valuation label: $\text{MOS} \geq 25 \rightarrow \text{UNDERVALUED}$ · $\text{MOS} < 0 \rightarrow \text{EXPENSIVE}$ · between → **FAIR** · no intrinsic value → **N/A**.

Per-stock DCF overrides (growth / discount / terminal) can be typed in the DCF Parameters table (§5.6). Overrides live only in your own browser session, reset when you leave, and never affect any other visitor.

3.3 Psychology Score

"Which way is the crowd leaning?" — computed from 3 months of price behaviour:

$$\begin{aligned} \text{Fear} &= (3\text{-month High} - \text{Price}) / \text{High} \times 100 \text{ [distance below the recent high]} \\ \text{Greed} &= \max((\text{Price} - \text{MA50}) / \text{MA50} \times 100, 0) \text{ [stretch above the 50-day average]} \\ \text{FOMO} &= \text{Greed} + \max(\text{weekly change}, 0) \\ \text{Psychology} &= \text{Fear} - \text{Greed} - \text{FOMO} \end{aligned}$$

Higher = fearful (contrarian-friendly); lower = greedy/overheated. The **Sentiment** pill reads directly off it: > 20 FEARFUL · > 5 CALM · < -20 OVERHEATED · < -5 GREEDY · else NEUTRAL.

3.4 Discovery Score

"Is the market noticing this stock?" — pure attention, deliberately containing no sentiment:

$$\text{Discovery} = \text{Activity} + \text{Volume Ratio} \times 10 + \text{Trend Score} + \text{News Score} + \text{Social Score}$$

where Activity = |weekly change|, Volume Ratio = latest volume ÷ 3-month average volume, Trend = Google search interest, News = NewsAPI + Yahoo headline scoring, Social = StockTwits message volume and bull/bear balance. In **attention-lite** contexts (overnight scans, live scans over 100 tickers, the featured card and the weekly digest) the Trend/News/Social terms are skipped and Discovery reflects price and volume only — always labelled where it applies.

3.5 The Long Score (0–100) and Investment Signal

The headline number. Every component is clamped to a fixed band first so no single factor can dominate:

$$\begin{aligned} \text{Long Score} &= \text{Quality} \times 0.35 + \text{MOS clamped to } \pm 50 \times 0.25 + \text{Psychology clamped to } \pm 50 \times 0.20 + \\ &\text{Discovery clamped to } 0\text{--}100 \times 0.20 \end{aligned}$$

Investment Signal: above 70 → **STRONG LONG** · above 50 → **LONG** · above 30 → **WATCHLIST** · otherwise **AVOID**. One override: if no intrinsic value could be computed (Valuation = N/A), the signal is **capped at WATCHLIST** — an unverifiable value leg cannot earn a full recommendation. The clamps affect only the score's arithmetic; tables always display the true unclamped MOS/Psychology/Discovery values.

3.6 Trade Setup — the timing axis

"Is right now a sane entry?" — deliberately separate from the Investment Signal. A great business can be a poor entry today; the site shows both verdicts rather than blending them.

Entry Zone = max(20-day support, MA50) – buy zone reached when $\text{Price} \leq \text{zone} \times 1.05$ Stop Loss
= support-based Targets: T1 = 20-day resistance, T2 = 60-day resistance, T3 = T2 × 1.10 (T3 shown only when Discovery > 50) Risk = Entry – Stop RRn = (Target n – Entry) / Risk

BUY requires all of: not in a confirmed downtrend, Psychology > 0, Discovery > 0, Price ≤ MA50 × 1.05, price near the Entry Zone, and RR1 ≥ 1.5. Otherwise WATCHLIST or AVOID. The **Trade Setup Score (0–100)** is a weighted display of the same gates (Trend Safety 20, Near Entry Zone 20, Risk/Reward 20, Price vs MA50 15, Psychology Momentum 12.5, Discovery Momentum 12.5) — the verdict as a number, not a separate formula.

3.7 The red-flag rule

Wherever a number rests on a **default or average because real data was unavailable**, it is rendered in **red** — in every table, on every page. A red Quality of 50 means “no fundamentals were available; this is the base assumption”, not a computed result. An estimate is never displayed as a fact.

4. Stock Deep Dive (/deep-dive)

Search one ticker to reach it. From top to bottom:

1. **Headline metrics:** Price, Intrinsic Value, MOS, Long Score, Signal.
2. **Watchlist button** (signed in): ☆ Add / ★ Remove this stock from your watchlist.
3. **Price chart:** the last 6 months of closes (teal), the 50-day moving average (dashed grey), and the Trade Filter's Entry Zone (dotted amber line).
4. **The plain-English case:** Why Buy / Why Wait / Risks in three columns, plus a suggested holding period — the same thesis engine used on Comparison, so words and numbers always agree.
5. **Price vs Intrinsic Value** bar chart (intrinsic bar green above price, red below) and the **Long Score gauge** with its threshold caption.
6. **“What's driving the Long Score”** — each factor's exact point contribution.
7. Below a divider (this is where the paywall gate will sit when subscriptions open): **Quality** (gauge + driving-terms chart, including the ROIC term), **Psychology** (gauge + Fear/Greed/FOMO chart, with the MA50 reading or a warning when it defaulted), **Discovery** (gauge + attention-terms chart), and the full **Trade Setup** (gauge, gate-check chart, entry/stop/targets bar chart, and a caption stating whether the price is currently inside the entry zone and the risk-reward per share).

5. Side-by-side Comparison (/comparison)

Search two or more tickers. Every stock is fetched and scored live, with a progress bar (a warning shows a realistic time estimate whenever more than 25 tickers were requested). All tables share one visual language:

- **Infill bars** — a value above a coloured bar whose fill width is the value and whose colour is its band: red / amber / green at the same thresholds the app verdicts on (Long Score 30/50, Quality 40/80, MOS 0/25, Psychology -5/+20, Discovery 25/50, Setup Score 45/65).
- **Verdict pills** — coloured labels: green = UNDERVALUED / FEARFUL / Uptrend / BUY / LONG / STRONG LONG; amber = FAIR / GREEDY / Ranging / WATCHLIST; red = EXPENSIVE / OVERHEATED / Downtrend / AVOID; grey = CALM / NEUTRAL / N/A. Type pills are neutral teal (categories, not verdicts) except red when the classification fell back to a default.
- **Price-relative colouring** — Intrinsic Value is green when above the current price (looks undervalued), red when below.
- **Red values** — the red-flag rule (§3.7) overrides all other colouring.

The tables, in order:

5.1 Comparison preview — Ticker, Price, Long Score (bar), Investment Signal (pill) for every stock scanned. This is the section that will remain free when subscriptions open; everything below it sits behind the (currently dormant) gate.

5.2 Side-by-side comparison — the main table, one row per ticker in the order you entered them. Columns: Ticker · Type (pill) · Price · **Intrinsic Value** (green/red vs price) · MOS (bar) · **Long Score** (bar) · **Quality Score** (bar — placed after Long Score) · Psychology (bar) · Discovery (bar) · Valuation (pill) · Sentiment (pill) · Trend (pill) · Trade Setup (pill) · Trade Setup Score (bar).

5.3 Average score by stock type — shown only when 5+ stocks were scanned (a group average of 2–3 names says nothing).

5.4 Top Investment Candidate — the highest Long Score among stocks with a verifiable valuation, with its full written thesis (Why Buy / Why Wait / Risks / holding period), followed by “Opportunity Details” expanders for the top five.

5.5 Trade Setup (collapsed expander) — the tactical entry layer for every scanned stock. Columns: Ticker · Trade Setup (pill) · Setup Score (bar) · Trend (pill) · **Entry Zone** — green when the price is currently at or inside the zone (a reachable entry now), red when the price is still above it (you’d be paying up; wait for the pullback) · Stop Loss · Targets 1–3 · Risk · **RR1/RR2/RR3** — coloured by *rank across the rows on screen*: the best risk/reward is green, the worst red, everything between amber (with a single stock: green ≥ 1.5 , amber ≥ 1.0 , else red) · Early Exit (amber “Yes” = watch flag). A “Position management & early-exit rules” expander beneath lists what to do as each target hits and when to abandon the position.

5.6 DCF Parameters (collapsed expander) — the valuation mechanics per stock: Ticker · Price · **Intrinsic Value** (green/red vs price, red when defaulted) · **IV/Price** (green above 1.00×, red below — e.g. 1.50× means the DCF values the stock at 1.5 times today’s price) · **Upside %** (the % gain from today’s price to the intrinsic-value target, green positive / red negative) · DCF Growth (red when defaulted) · Growth Governor (which rule set the growth figure) · Discount · Perpetual · Long Score (bar). Click **Enable manual override** to type your own growth/discount/terminal per stock — blank = automatic; Save, then search again to recompute. Overrides are session-only.

(The former “Full Stock Database” table has been removed — its per-column detail lives in each stock’s Deep Dive.)

A closing line reports how many stocks were analysed and how long the scan took. If some interaction re-renders the page without a new scan, a caption says the results are the last completed scan, not fresh prices.

6. Stock Scanner (/scanner)

Answers the bigger question: “across a whole index, what should I even look at?”

Selection. Tick Australia and/or USA, then pick exactly one universe — each maps to one real index, never a blend:

Universe	Constituents sourced from	Fallback if the source is down
ASX 200	Wikipedia S&P/ASX 200 (live, cached 24h)	Curated local ASX list
ASX 300	asx300list.com (live)	ASX 200, then the local list
S&P 500	Wikipedia (live)	Static 10-ticker list
Nasdaq 100	Wikipedia (live)	—
Russell 2000	iShares IWM holdings CSV (live)	Last successfully fetched list, saved to disk
Small Caps (S&P 600)	Wikipedia (live)	—

The source actually used is always shown under the pickers, so a scan on fallback data is never silently presented as live.

Sector filter with heat dots. The Sector dropdown decorates each sector with a coloured dot:  hot /  medium /  cold, ranked by dollar-volume-weighted 12-month return **relative to the other sectors in this universe** (top third green, bottom third red). Computing it for a universe the first time takes a few minutes (a year of prices for a sample of each sector); after that it is cached for the day. Country mood lines (§2) appear above when available.

Overnight scans — the instant path. Every night at 6:00am Brisbane the server scans each configured universe by itself and stores the ranked result. When one exists for the selected universe, an expander appears: “Overnight ASX 200 scan — 197 stocks ranked by Long Score, computed ...”. Inside is the full colour-coded table (same bars/pills as §5): Ticker · Type · Price · Intrinsic Value (vs price) · MOS · Long Score · Quality · Psychology · Discovery · Valuation · Signal · Trend · Trade Setup. Overnight scans are attention-lite (§3.4), labelled as such, and marked with their computation time; results older than 3 days are treated as expired. If a night's run is interrupted (e.g. a redeploy), it retries automatically, up to three attempts per day.

Live scan. The Run Scan button scores the selected universe/sector live, with a progress bar and an honest up-front time estimate. Scans over 100 tickers automatically run attention-lite to protect speed and shared API quotas (stated in the warning). Results render in exactly the layout of §5.

7. Rational Compounder Analysis (/research)

The site's hand-built research — not computed, written. Every chart, threshold and colour band comes from the author's own research workbook. A “Last updated” badge shows when the underlying data snapshot was rebuilt.

Pick a **Stock** (currently AUB.AX, CSL.AX, RMD.AX — new companies are added as each one's research completes) and a **Section**:

- **Fundamentals** — ~10 years of monthly share price against the author's own 10-year average line, colour-banded metric gauges (each with a collapsed "What this measures" explanation in the author's words), other headline metrics as cards, and a share-price-growth-by-year chart.
 - **Value vs Book** — $\text{Intrinsic Value} \div \text{Book Value}$ by year modelled, colour-banded, with the average line.
 - **Dividends** — the "value created per \$ retained" test at four horizons (2Y/5Y/10Y/TTM): for every dollar of earnings kept, how much market value was created — plus dividend metrics.
 - **Earnings Trends** — EPS by year, EPS growth by year (green/red), P/E by year with 3-year-average and overall-average reference lines, plus the sheet's other figures.
 - **Cost of Capital** — WACC vs ROIC by period: the fundamental test of whether the business earns more on its capital than that capital costs.
 - **Fair Value** (*gated when subscriptions open*) — four independent valuation methods side by side against the current price: trailing P/E, forward P/E, DCF, and the 10-year equity method, with the exact inputs behind each.
 - **Company Potential** (*gated when subscriptions open*) — the author's Buffett/Munger-style judgment: Low/Medium/High ratings on management, moat, risk, pricing power and more (green/amber/red pills), quick yes/no checks, and the full written analysis grouped by theme (the business & its moat, risk & inversion, psychology, the investment case, financial diligence, management & context).
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8. Colour reference (one table for the whole site)

Element	Green	Amber	Red	Grey
Long Score bar	> 50	30–50	≤ 30	—
Quality bar	> 80	40–80	≤ 40	—
MOS bar	> 25%	0–25%	≤ 0%	N/A
Psychology bar	> 20	–5 to 20	≤ –5	—
Discovery bar	> 50	25–50	≤ 25	—
Setup Score bar	> 65	45–65	≤ 45	—
Valuation pill	UNDERVALUED	FAIR	EXPENSIVE	N/A
Sentiment pill	FEARFUL	GREEDY	OVERHEATED	CALM / NEUTRAL
Trend pill	Uptrend	Ranging	Downtrend	—
Signal pills	BUY / LONG / STRONG LONG	WATCHLIST	AVOID	—
Intrinsic value	above current price	—	below current price	N/A
IV/Price	> 1.00×	—	< 1.00×	N/A
Upside %	positive	—	negative	N/A
Entry Zone	price at/inside zone	—	price above zone	—
RR1/RR2/RR3	best on screen	in between	worst on screen	no value
Any value in red bold	—	—	estimated/default input (red-flag rule)	—

9. Accounts, watchlist and the weekly digest

Signing in (Google, top right) requires no password creation; the site receives only your name and email. It unlocks: the **watchlist** (☆ button on any Deep Dive; your saved stocks appear as one-click chips on the home page), feedback attributed to your email, and the **weekly digest**.

The weekly digest arrives Monday mornings (Brisbane time) if your watchlist has stocks in it: each saved stock re-scored (attention-lite) with price, intrinsic value, MOS, Long Score and signal, each ticker linking to its live Deep Dive. To stop receiving it, remove all stocks from your watchlist.

Feedback — the “Tell us what you think” button on any results page goes straight to the author’s inbox, or email rationalcompounder@stocksdeepdive.com.

Subscriptions are not yet active. The site is free during launch; when subscriptions open (Stripe), the sections marked *gated* above become subscriber content, previews stay free, and founding members keep launch pricing.

10. Operations reference (site owner)

Everything below is configured with Railway environment variables; changing a variable restarts the service (never do this mid-scan), and after any restart the background scheduler starts on the first page visit.

Variable	Default	Purpose
NIGHTLY_UNIVERSES	ASX 200	Comma-separated universes to pre-scan nightly (names exactly as in §6)
NIGHTLY_SCAN_UTC_HOUR	20	Scan hour, UTC (20 = 6am Brisbane). Scans retry $\leq 3\times$ /day if a run was interrupted
DIGEST_UTC_WEEKDAY / DIGEST_UTC_HOUR	6 / 21	Digest schedule (Sunday 21:00 UTC $\approx$ Monday 7am Brisbane)
DIGEST_FORCE	—	Set to any <i>new</i> value to send the digest immediately, once per value
SCHEDULER_ENABLED	true	false disables both background jobs
MAILGUN_API_KEY / MAILGUN_DOMAIN	—	Email sending (shared with the feedback engine)
NEWS_API_KEY	—	Optional; enriches News Score. Without it, Yahoo headlines only
ADMIN_REFRESH_KEY	—	Enables the ☰ admin panel on the Research page (workbook upload → rebuild)
PAYWALL_ENABLED	off	The subscription master switch (needs Stripe + Google OAuth vars; see <code>paywall_engine.py</code>)

Persistent data (watchlists, overnight scans, compounder data, scheduler state) lives on the attached Railway Volume and survives redeployments. Manual commands from the service shell: `python nightly_scan.py "S&P 500"` (one universe now) and `python digest_engine.py` (send digest now).

11. Data sources and honest limitations

Prices, fundamentals, and analyst estimates come from Yahoo Finance; search interest from Google Trends; headlines from NewsAPI and Yahoo; social chatter from StockTwits; country mood from GDELT. These are free public feeds: data can be delayed, revised, or occasionally missing — which is exactly what the red-flag rule exists to expose rather than hide. Intrinsic values are estimates resting on stated assumptions. Nothing on the site considers any individual's circumstances, and nothing in it is financial advice — see the disclaimer in the site footer on every page.